

Anomaly Detection as an Inverse Problem on Quotient Spaces

Equivalence Relations and Functional Manifolds as a General Diagnostic Framework
Technical Note — Precedence Record

Eduardo Di Santi¹

¹College of Engineering and Applied Science, University of Colorado Boulder

June 28, 2026

Abstract

This work proposes a quotient-space formulation of diagnostic monitoring in physical systems. The central claim is that anomaly detection is not fundamentally a supervised classification problem in raw observation space, nor does it require identifying the full forward physical model, which may be unknown, nonlinear, high-dimensional, or impractical to learn. Instead, the objective is to recover a representation in which observations that are physically equivalent with respect to the latent state are mapped to the same equivalence class.

The key step is to separate physical-state information from nuisance variability. Measurements are affected by transformations such as amplitude scaling, phase shift, speed variation, sensor offset, installation effects, and additive noise. These transformations modify the observed signal while preserving the underlying physical state. They therefore induce an equivalence relation on the observation space. The resulting quotient space is the natural geometric domain for diagnosis: the inverse problem is no longer posed as explicit recovery of the full generative model, but as recovery of membership in physically meaningful equivalence classes.

Under this formulation, a nominal operating condition is represented not by a single waveform or label, but by a connected functional manifold in the quotient space. Anomaly detection becomes the problem of deciding whether an observation belongs to this nominal quotient manifold. Fault classification, when required, becomes a secondary operation: once an observation is detected outside the nominal manifold, it may be assigned to another quotient-space region or fault manifold. This provides a geometric mechanism for the emergence of functional manifolds: they arise by collapsing nuisance orbits that are irrelevant to the physical state.

After establishing the mathematical object, we introduce one computational approximation. Nominal trajectories are generated by Monte Carlo simulation over nuisance variables, a Conv1D autoencoder is trained only on the synthetic nominal orbit, and the reconstruction error is used as an inverse-problem residual. An adaptive threshold estimates the local boundary of the nominal manifold for each physical asset. Synthetic validation on four parallel bearing signals provides preliminary evidence that quotient-space recovery is operationally feasible under controlled conditions. The experiments do not prove the mathematical formulation itself; they demonstrate that one computational approximation can recover sufficient quotient geometry to support anomaly detection and subsequent signal classification.

1 Introduction

Diagnostic monitoring in physical systems is often reduced to a pattern recognition problem: given an observation window, assign a semantic label such as nominal, degraded, or faulty. In rotating machinery, this usually becomes a bearing fault classification task, where vibration windows are mapped to labels such as healthy operation, outer-race fault, inner-race fault, or ball fault. This formulation is operationally convenient, but it hides a more fundamental structure.

The physical state of a system is not observed directly. Instead, one observes a signal produced by a measurement process that mixes the latent physical state with nuisance factors such as operating speed, amplitude, phase, sensor gain, installation conditions, measurement offsets, and noise. Consequently, two observations can look different in raw signal space while representing the same physical condition. Conversely, two observations can appear superficially similar while belonging to different physical states.

This is the signature of an inverse problem. The observation is not the object of interest; it is the output of a forward process. However, in many industrial settings the full forward model may be unknown, nonlinear, high-dimensional, or impractical to identify. The objective is therefore not necessarily to learn or invert the complete generative model (F). The objective is to recover a representation in which physically equivalent observations are identified, while physically meaningful deviations remain detectable.

The central claim of this work is the following:

Diagnostic monitoring can be formulated as inverse recovery over a quotient space induced by nuisance transformations of the observation process.

The quotient space appears because nuisance transformations modify the observed signal without changing the underlying physical state. These transformations induce an equivalence relation on the observation space. The diagnostic task can therefore be reformulated as recovery of membership in physically meaningful equivalence classes, rather than direct classification in raw observation space or explicit reconstruction of the full forward physical model.

Under this view, a nominal operating condition is not represented by a single waveform or by a predefined label. It is represented by a connected functional manifold in quotient space. Anomaly detection is then a membership test: an observation is nominal if it belongs to the nominal quotient manifold and anomalous if it does not. Classification, when required, becomes a secondary operation: once an observation is detected outside the nominal manifold, it may be assigned to another quotient-space region or fault manifold.

The quotient space is not introduced because an autoencoder is used. The autoencoder is introduced because the diagnostic problem naturally lives on a quotient space. This distinction is essential. The mathematical formulation should remain valid if the computational operator is replaced by another representation learner, such as a contrastive encoder, a predictive latent model, a diffusion-based score model, or a physics-informed residual operator.

This work separates three layers that are often conflated:

- (i) the physical observation model;
- (ii) the quotient-space inverse problem induced by nuisance equivalence;
- (iii) one practical computational approximation.

The first two layers define the scientific contribution. The third layer demonstrates that the formulation can be approximated in a concrete setting. In this work, the concrete setting is a

synthetic bearing fault detection experiment, used not as the center of the theory but as a controlled validation of the proposed quotient-space inverse recovery framework.

2 Diagnostic Monitoring as Inverse Recovery

Equation (??) describes a forward observation map. The inverse diagnostic problem is ill-posed in the classical sense: multiple nuisance configurations may produce different observations corresponding to the same latent physical state. The goal is therefore not to invert F exactly, nor to identify all nuisance variables explicitly. Instead, the goal is to construct a diagnostic representation

$$\Phi : \mathcal{X} \rightarrow \mathcal{Z} \quad (1)$$

such that $\Phi(x)$ preserves information about the latent physical state while suppressing nuisance variability.

A direct classifier attempts to approximate the map

$$x \mapsto y,$$

where y is a semantic or operational label. However, the label is not the primitive object. The primitive object is the latent physical state θ , and semantic labels are assigned only after a stable diagnostic structure is recovered.

This motivates a two-step view of diagnostic monitoring:

- (a) recover the geometric structure associated with latent-state equivalence classes;
- (b) assign operational semantics to the recovered structures.

In this view, classification is downstream of inverse recovery. Anomaly detection is not the prediction of a predefined class label. It is the decision that an observation does not belong to the recovered nominal structure. If additional non-nominal structures are recovered, subsequent classification may be performed by assigning the observation to another quotient-space region or fault manifold.

The experimental bearing case studied later in this work should therefore be read as one concrete instantiation of this general formulation. The theory concerns diagnostic monitoring under nuisance transformations; the bearing simulator provides a controlled physical example in which the proposed quotient-space recovery can be evaluated.

3 Quotient-Space Formulation

The key mathematical object is the equivalence relation induced by nuisance transformations. Let G_{nuis} denote a group, or approximate transformation family, acting on $\mathcal{X}$. For $g \in G_{\text{nuis}}$, the transformed observation $g \cdot x$ represents a change in the measured signal that does not alter the underlying latent physical state.

Definition 1 (Nuisance equivalence). *Two observations $x_1, x_2 \in \mathcal{X}$ are nuisance-equivalent, written $x_1 \sim x_2$, if there exists $g \in G_{\text{nuis}}$ such that*

$$x_2 = g \cdot x_1, \quad (2)$$

and both observations correspond to the same latent physical state θ .

The orbit of an observation x under nuisance transformations is

$$[x] = \{g \cdot x : g \in G_{\text{nuis}}\}. \quad (3)$$

The quotient space is the set of such equivalence classes:

$$\mathcal{X}/G_{\text{nuis}} = \{[x] : x \in \mathcal{X}\}. \quad (4)$$

Proposition 1 (Geometric formulation of diagnostic monitoring). *Assume that the nuisance transformations preserve the latent physical state. That is, for observations generated by $x = F(\theta, \eta)$, the action of $g \in G_{\text{nuis}}$ modifies nuisance degrees of freedom but not θ . Then inverse recovery of the latent physical state depends only on the equivalence class $[x]$, not on the particular representative x . Consequently, the natural domain of diagnostic monitoring is the quotient space $\mathcal{X}/G_{\text{nuis}}$, rather than the raw observation space $\mathcal{X}$.*

Proof. Let $x_1, x_2 \in \mathcal{X}$ satisfy $x_2 = g \cdot x_1$ for some $g \in G_{\text{nuis}}$. By assumption, g changes only nuisance variables and preserves the latent physical state. Thus both observations correspond to the same θ . Any diagnostic map that recovers information about θ must therefore assign the same physical-state information to x_1 and x_2 . Hence the diagnostic map is constant on nuisance orbits and factors through the quotient projection

$$\pi : \mathcal{X} \rightarrow \mathcal{X}/G_{\text{nuis}}.$$

Therefore diagnostic monitoring is naturally posed on the quotient space. $\square$

Corollary 1 (Equivalence-class diagnosis). *If two observations lie in the same nuisance orbit, no physically meaningful diagnostic decision should distinguish them on the basis of nuisance variation alone. Consequently, nominal observations should be identified as elements of a nominal equivalence class, or a connected region of such classes, in the quotient space.*

Definition 2 (Nominal quotient manifold). *The nominal quotient manifold $\mathcal{M}_{\text{N}}$ is the connected subset of $\mathcal{X}/G_{\text{nuis}}$ generated by observations sharing the nominal latent physical state:*

$$\mathcal{M}_{\text{N}} = \{[x] \in \mathcal{X}/G_{\text{nuis}} : x = F(\theta_{\text{N}}, \eta), \eta \in \mathcal{E}\}. \quad (5)$$

Under this definition, anomaly detection becomes a membership problem:

$$[x] \in \mathcal{M}_{\text{N}} \quad \text{or} \quad [x] \notin \mathcal{M}_{\text{N}}. \quad (6)$$

This is the point at which anomaly detection enters the formulation. An anomaly is not primarily a label. It is a failure of membership in the nominal quotient manifold. Classification, when required, is a subsequent step that assigns non-nominal observations to other quotient-space regions or fault manifolds.

4 Functional Manifolds as Quotient Geometry

The quotient-space formulation provides a geometric mechanism through which functional manifolds emerge. A functional manifold is not postulated as an arbitrary structure discovered by an algorithm. It arises because nuisance transformations generate orbits in observation space, and the quotient projection collapses each orbit into a single equivalence class.

Thus, the observed manifold structure has two sources:

- (i) physical continuity of the underlying system;

- (ii) identification of observations that differ only by nuisance transformations.

In raw signal space, nominal operation may appear as a broad, noisy cloud. In quotient space, the nuisance directions are collapsed. The remaining geometry is closer to the latent physical-state structure. This is why learning a representation that is invariant to nuisance transformations can reveal compact functional manifolds.

This mechanism is illustrated conceptually in Figure 1. The quotient projection does not create physical structure *ex nihilo*; it removes directions of variability that are irrelevant to the latent state. What remains is a lower-complexity geometry in which nominal and non-nominal regimes can be represented as functional manifolds.

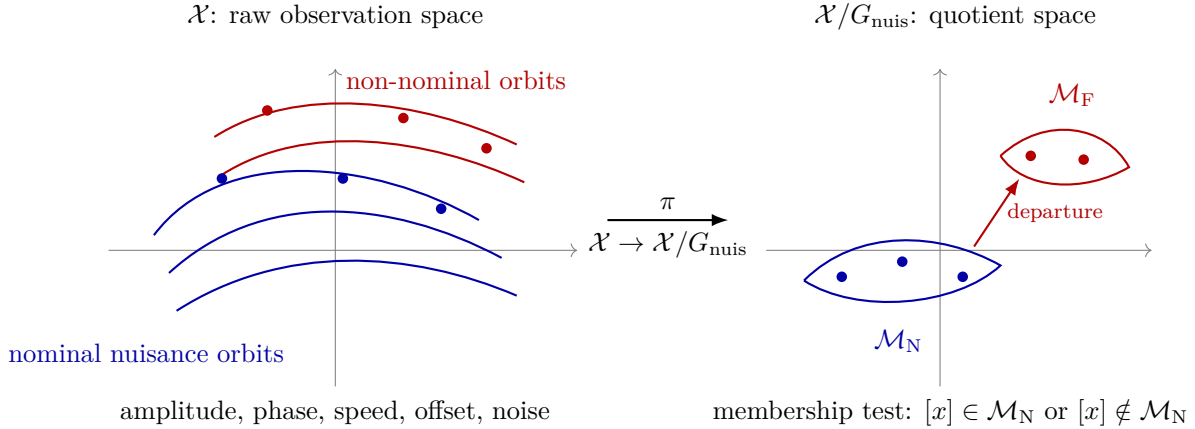


Figure 1: Conceptual illustration of functional manifolds as quotient geometry. In raw observation space, nuisance transformations generate extended orbits of observations that preserve the same latent physical state. The quotient projection π collapses these orbits into equivalence classes, revealing compact nominal and non-nominal functional manifolds.

This also clarifies the relation between anomaly detection and classification. A classifier learns a partition of labeled observations. A quotient-space diagnostic operator learns a representation in which physically equivalent observations are identified. Operational labels may then be attached to the recovered structures, but the structures themselves precede the labels.

5 One Computational Approximation

The previous sections establish the mathematical object. We now introduce one computational approximation.

The approximation has four components:

- (1) Monte Carlo generation of nominal trajectories over nuisance variables;
- (2) self-supervised representation learning on nominal trajectories only;
- (3) reconstruction error as an inverse-problem residual;
- (4) per-asset adaptive threshold estimation.

This section should be interpreted as one realization of the quotient-space formulation, not as the formulation itself. The same mathematical framework could be implemented with alternative representation operators, residual functions, or domain simulators.

5.1 Computational Strategy

Let θ_N denote a nominal latent physical state. The computational objective is to approximate the nominal quotient manifold $\mathcal{M}_N$ without learning the full forward model F . Instead of identifying all physical parameters and nuisance variables explicitly, we generate nominal observations under nuisance variability and train a representation operator that is stable on this nominal orbit.

Synthetic nominal samples are generated as

$$x_i = F(\theta_N, \eta_i), \quad i = 1, \dots, N, \quad (7)$$

where each $\eta_i \in \mathcal{E}$ represents a nuisance configuration. These samples approximate the orbit of the nominal state under nuisance transformations.

The purpose of Monte Carlo generation is not to model every possible non-nominal condition. The purpose is to sample the nominal nuisance orbit densely enough that a representation learner can approximate the quotient geometry of nominal operation. Non-nominal regimes are then detected by their failure to belong to the recovered nominal structure.

5.2 Controlled Bearing Instantiation

The controlled validation uses a synthetic bearing signal simulator designed to expose the geometric structure of the proposed formulation rather than to reproduce a specific industrial asset. In this instantiation, the generic observation $x \in \mathcal{X}$ is a vibration window. Each window has length $n = 1200$, corresponding to 0.1 seconds sampled at $f_s = 12,000$ Hz. The nominal shaft frequency is 30 Hz.

The nominal bearing signal is generated as a finite harmonic expansion of the shaft frequency:

$$x_N(t) = \sum_{k=1}^5 a \rho^{k-1} \sin(2\pi k f_0 s t + \phi/k),$$

where a is an amplitude factor, $\rho = 0.55$ is the harmonic decay, $f_0 = 30$ Hz is the nominal shaft frequency, s is a speed-scaling factor, and ϕ is a phase perturbation.

The nuisance variables are sampled by Monte Carlo simulation:

$$a \sim U(0.85, 1.15), \quad \phi \sim U(-0.3, 0.3), \quad s \sim U(0.95, 1.05),$$

together with additive offset, additive noise, and a small local dropout perturbation. These variables modify the observed waveform without changing the intended physical regime. They therefore instantiate the nuisance orbit discussed in the quotient-space formulation.

Non-nominal regimes are generated by adding impulsive components to the nominal base signal. The outer-race fault uses an impulse train near 108s Hz, the inner-race fault uses an impulse train near 162s Hz, and the ball-fault regime uses an impulse train near 72s Hz with a low-frequency amplitude modulation. These non-nominal regimes are not used to train the representation operator; they are reserved for evaluating whether departure from the nominal quotient manifold can be detected and subsequently classified.

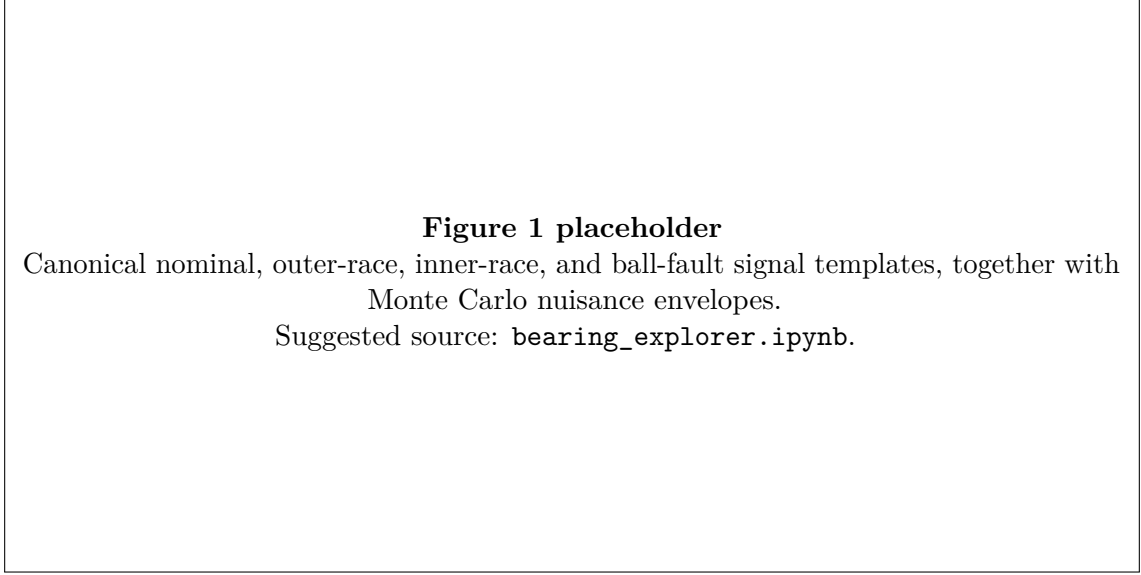


Figure 2: Controlled bearing signal instantiation. The nominal signal and the non-nominal fault regimes are generated from distinct physical templates, while nuisance transformations modify the observed waveform without changing the intended latent regime.

5.3 Representation Operator

Let

$$E : \mathcal{X} \rightarrow \mathcal{Z}$$

be an encoder and

$$D : \mathcal{Z} \rightarrow \mathcal{X}$$

a decoder. In the present implementation, E and D are realised by a Conv1D autoencoder trained exclusively on nominal synthetic windows.

The learned reconstruction is

$$\hat{x} = D(E(x)). \quad (8)$$

The reconstruction residual is

$$r(x) = \frac{1}{n} \|x - \hat{x}\|_2^2. \quad (9)$$

The residual is interpreted as a computable proxy for distance to the learned nominal structure:

$$r(x) \approx d([x], \mathcal{M}_N). \quad (10)$$

This interpretation is important. The contribution is not the use of mean-squared error as such. The contribution is the use of a residual as an operational test for membership in the nominal quotient manifold. In this implementation, the residual is a reconstruction error; in another implementation, it could be replaced by a latent likelihood, an energy score, a predictive error, or a physics-informed residual.

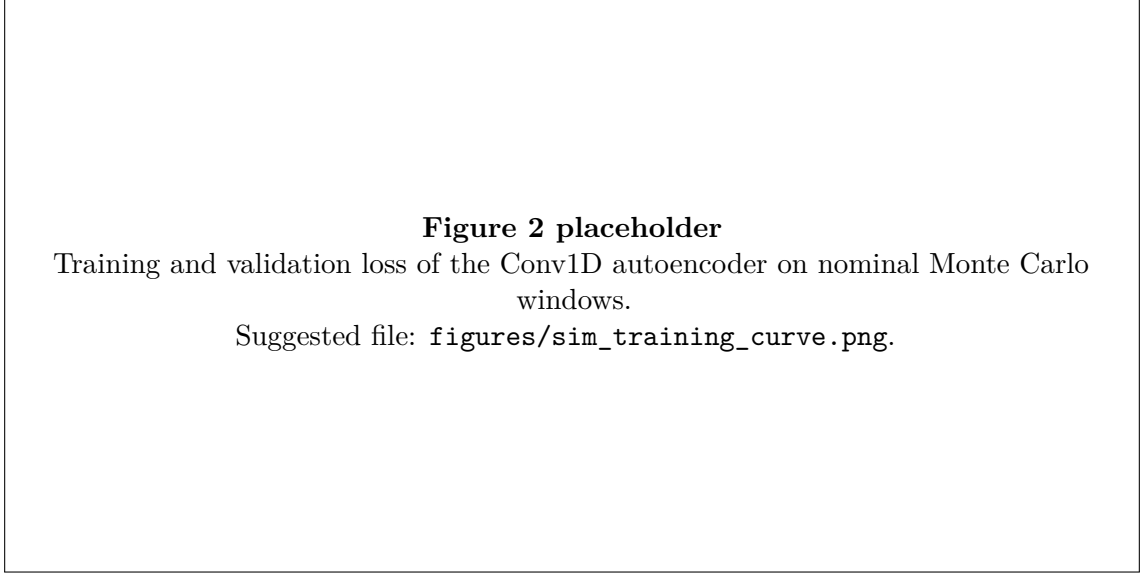


Figure 3: Self-supervised learning of a nominal reconstruction operator from synthetic nuisance-orbit samples. The learned operator is used as one computational approximation of the nominal quotient geometry.

5.4 Monte Carlo Calibration

A Monte Carlo reference threshold is estimated from residuals on held-out nominal synthetic samples. Let

$$\mu_{\text{MC}} = \mathbb{E}_{\text{MC}}[r(x)], \quad \sigma_{\text{MC}} = \text{Std}_{\text{MC}}[r(x)].$$

More generally, let $q \in (0, 1)$ denote an operational quantile controlling the trade-off between sensitivity and false-alarm tolerance. A parametric calibration can be written as

$$\tau_{\text{MC}}(q) = \mu_{\text{MC}} + z_q \sigma_{\text{MC}}, \tag{11}$$

where z_q is the corresponding standard-normal quantile or an empirically estimated equivalent. Alternatively, $\tau_{\text{MC}}(q)$ can be estimated directly as the empirical q -quantile of the nominal residual distribution.

The value of q is not part of the mathematical formulation. It is an operational risk parameter. Lower values increase sensitivity but may increase false alarms; higher values reduce false alarms but may delay detection of weak or incipient deviations.

In the synthetic experiment reported below, q is treated as a calibration parameter. The resulting $\tau_{\text{MC}}(q)$ defines a reference boundary for the learned nominal structure.

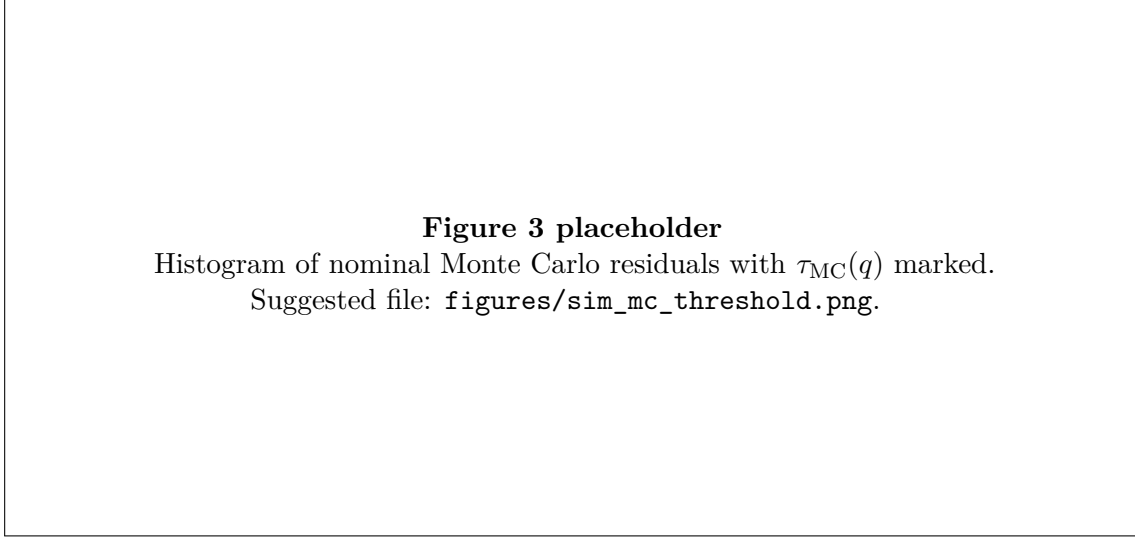


Figure 4: Monte Carlo calibration of the nominal residual distribution. The threshold defines a synthetic reference boundary for the learned nominal quotient structure.

5.5 Per-Asset Adaptive Threshold

A global threshold is insufficient when multiple physical assets have different local baselines. Even if they share the same representation operator, each asset may have a distinct residual scale due to installation, operating conditions, sensor coupling, or local environmental effects.

For each asset b , an adaptive threshold is estimated from a sliding buffer of recent non-anomalous residuals:

$$\tau_t^{(b)}(q) = \mu_t^{(b)} + z_q \sigma_t^{(b)}. \quad (12)$$

Here $\mu_t^{(b)}$ and $\sigma_t^{(b)}$ are computed locally for asset b . The scalar z_q preserves the statistical meaning of the selected operational risk level, while the local mean and standard deviation adapt the threshold to each physical asset.

The anomaly decision is

$$a_t^{(b)} = \begin{cases} 1, & r(x_t^{(b)}) > \tau_t^{(b)}(q), \\ 0, & r(x_t^{(b)}) \leq \tau_t^{(b)}(q). \end{cases} \quad (13)$$

This separates the representation problem from the operational decision problem. The representation is shared across assets. The decision boundary is local. In the bearing experiment, each physical asset b corresponds to one simulated bearing; in the general formulation, b may represent any monitored component, subsystem, or sensorized physical unit.

6 Synthetic Validation

The objective of the synthetic experiment is not to prove the quotient-space formulation. The proposition in Section 4 is a structural statement about the geometry of the diagnosis problem under nuisance transformations. The experiment tests whether the proposed computational approximation can recover that geometry sufficiently well to separate healthy and faulty trajectories.

Four simultaneous synthetic bearings are simulated:

- Bearing A: always healthy;

- Bearing B: outer-race fault onset;
- Bearing C: inner-race fault onset;
- Bearing D: ball fault onset.

The representation operator is trained only on healthy synthetic data. Fault data are used only for evaluation. The adaptive threshold uses a sliding window of 200 samples. The Monte Carlo calibration parameter is $z_{MC} = 2.79$.

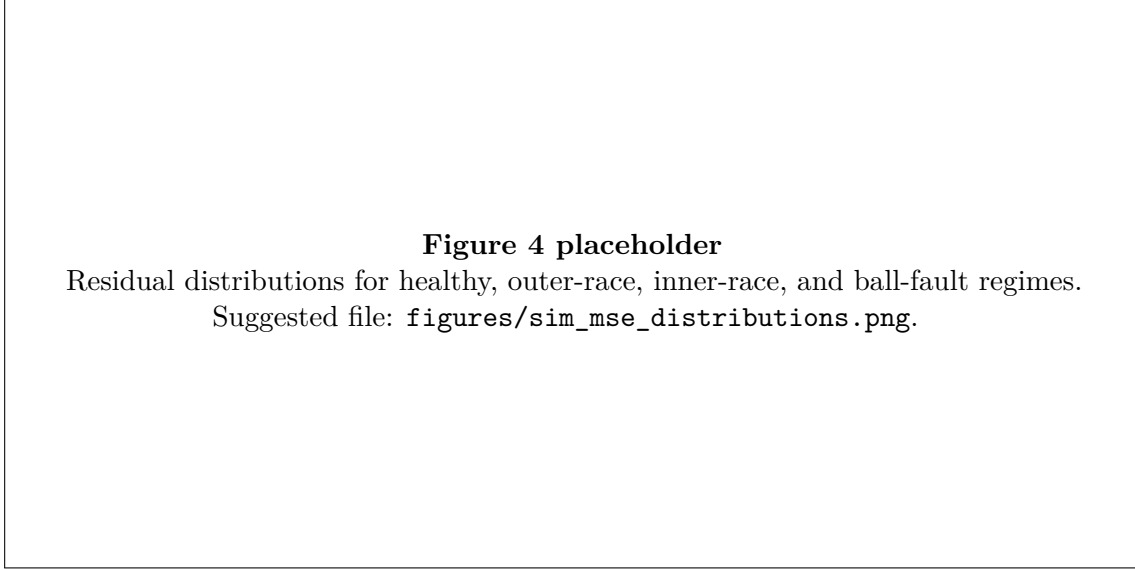


Figure 5: Residual distributions produced by the learned representation operator. Clear separation between healthy and faulty regimes indicates that the approximation recovers diagnostically meaningful quotient geometry.

Table 1: Synthetic per-bearing detection metrics. FAR denotes false-alarm rate during healthy operation; DR denotes detection rate after fault onset.

Bearing	Scenario	FAR during healthy period	DR during fault period
A	Always healthy	1.5%	N/A
B	Outer-race fault	2.0%	100%
C	Inner-race fault	2.0%	100%
D	Ball fault	2.0%	86%

The results show that the learned residual remains stable for the healthy bearing and increases after fault onset for the faulty bearings. The outer-race and inner-race fault regimes are detected completely in the evaluation interval. The ball-fault regime is detected less strongly, with an 86% detection rate, which is consistent with a weaker or less separable deviation from the learned healthy structure.

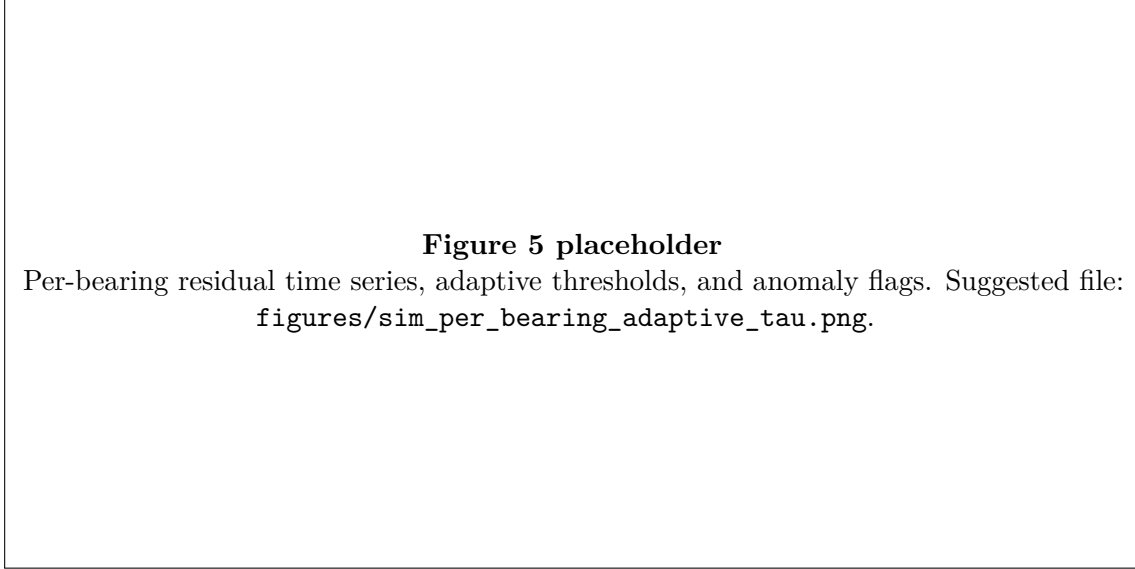


Figure 6: Per-bearing adaptive thresholds. Each bearing estimates its own local boundary of healthy operation, preventing cross-unit contamination.

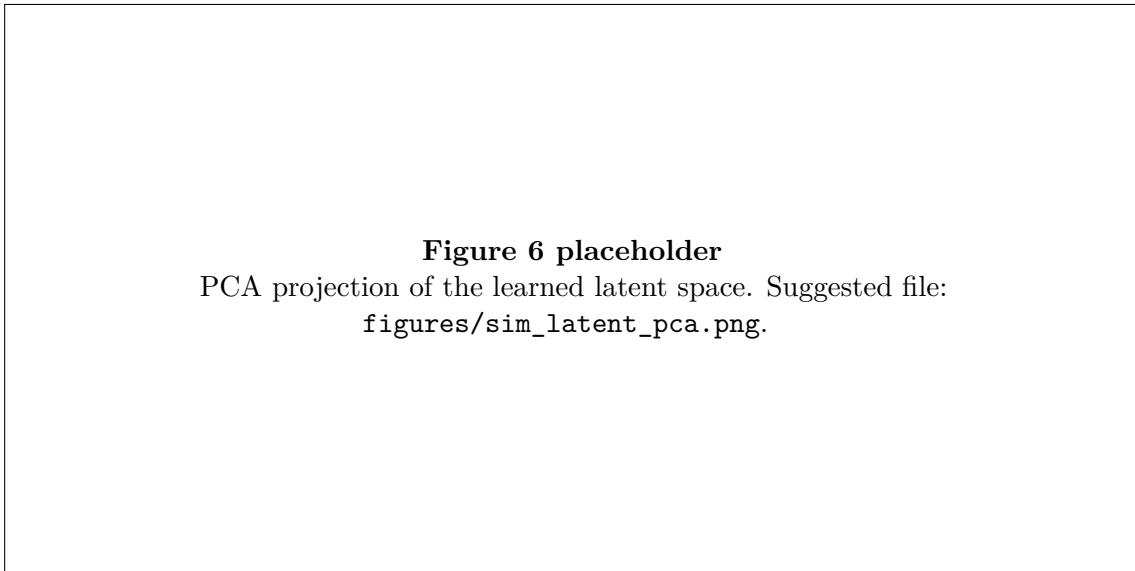


Figure 7: Latent-space visualization of the learned representation. The plot should be interpreted as evidence of geometric separation, not as the mathematical definition of the quotient space itself.

7 Interpretation

The experiment supports three claims.

First, a representation trained only on healthy nuisance-orbit samples can produce residuals that distinguish healthy and faulty regimes. This is consistent with the interpretation of the residual as a proxy for distance to the healthy quotient manifold.

Second, the per-bearing adaptive threshold is necessary. A shared representation can be global,

but the alarm boundary must be local to the asset. This reflects the distinction between learning a common quotient geometry and estimating a local operational boundary.

Third, the fault classes need not be learned as semantic labels. The outer-race, inner-race, and ball-fault regimes are detected by departure from the healthy structure. This is exactly the inverse-problem view: first recover membership in the healthy functional structure; assign semantics later.

8 Discussion

The proposed formulation differs from standard supervised fault classification in three ways.

Diagnosis precedes classification. The primary decision is whether an observation belongs to the healthy quotient manifold. Classification of the type of fault is a subsequent semantic step.

The representation is geometric. The goal is not to memorize labels, but to collapse nuisance orbits while preserving physical health-state information. This is why the quotient space is the natural object.

The threshold is asset-local. A physical asset may have a local residual baseline even when the representation operator is shared. Per-bearing adaptive thresholding estimates this local boundary without retraining the representation.

This note also clarifies the relationship between quotient spaces and functional manifolds. The quotient-space formulation provides the geometric mechanism through which functional manifolds emerge. They are connected regions of the quotient space after nuisance directions have been collapsed. This gives a mathematical origin to the compact structures observed in self-supervised diagnostic representations.

9 Limitations

This note is a precedence record and should not be read as a complete industrial validation.

The current experiment is synthetic. It demonstrates the feasibility of the computational approximation under controlled conditions, but it does not establish robustness under field conditions.

The nuisance transformation family is approximate. In real machines, nuisance effects may not form an exact mathematical group. Nevertheless, the quotient-space formulation remains useful as an organizing principle when transformations approximately preserve the physical state.

The autoencoder is only one possible approximation. Future work should compare reconstruction-based residuals with alternative representation operators and distance functions.

Finally, the detection of fault type is not fully addressed here. The present note focuses on anomaly detection as failure of membership in the healthy quotient manifold. Fault classification can be formulated as identifying the destination manifold after departure from the healthy structure.

10 Conclusion

This work establishes a mathematical formulation of bearing diagnosis as inverse recovery over quotient-space geometry. The formulation begins with the physical observation model, identifies

nuisance transformations that preserve the health state, and shows that diagnosis naturally factors through the quotient space induced by those transformations.

The proposed learning architecture constitutes one practical approximation of that formulation. Monte Carlo simulation samples the healthy nuisance orbit, a self-supervised representation operator approximates the healthy quotient geometry, and a per-bearing adaptive threshold estimates the local decision boundary. Synthetic experiments provide preliminary empirical evidence that quotient-space recovery is feasible under controlled operating conditions.

The main contribution is therefore not a particular autoencoder architecture. The main contribution is the geometric formulation: bearing anomaly detection is a membership problem on a quotient-space functional manifold induced by nuisance-invariant observations.

Future work will investigate field validation, alternative representation operators, fault-manifold classification after anomaly detection, and transfer to industrial rotating machinery datasets.

References

- [1] J. Hadamard. *Lectures on Cauchy's Problem in Linear Partial Differential Equations*. Yale University Press, 1923.
- [2] A. N. Tikhonov and V. Y. Arsenin. *Solutions of Ill-Posed Problems*. Winston, 1977.
- [3] T. Kohonen. Self-organized formation of topologically correct feature maps. *Biological Cybernetics*, 43, 59–69, 1982.
- [4] W. A. Smith and R. B. Randall. Rolling element bearing diagnostics using the Case Western Reserve University data: A benchmark study. *Mechanical Systems and Signal Processing*, 64–65, 100–131, 2015.
- [5] E. Di Santi. *Self-Supervision, Manifold Saturation, and Emergent Operational Classes: Learning from Scarce Observations as Inverse Recovery of Compact Functional Structure*. Technical report, University of Colorado Boulder, 2026.